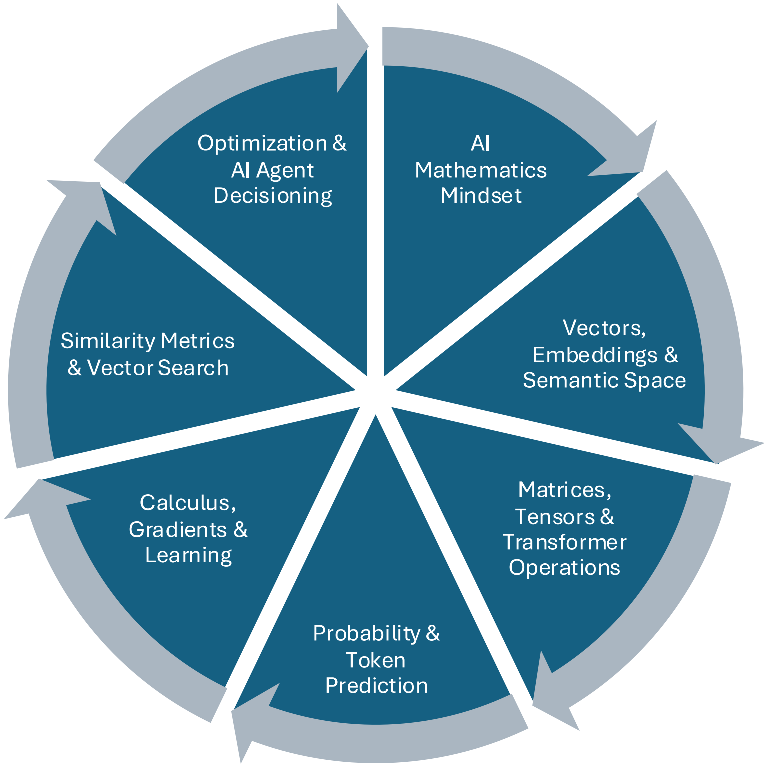
Mathematical Foundation for AI Systems

Session Objective

- By the end of the session, learners should:
  - Understand the mathematics behind AI systems in simple practical terms
  - Relate math concepts directly to ML, Deep Learning, LLMs, and Agentic AI
  - Perform basic mathematical operations used inside AI pipelines
  - Develop intuition rather than theoretical memorization

Spread



AI Mathematics Mindset
Vectors, Embeddings & Semantic Space
Matrices, Tensors & Transformer Operations
Probability & Token Prediction
Calculus, Gradients & Learning
Similarity Metrics & Vector Search
Optimization & AI Agent Decisioning

AI Mathematics Mindset

Permutation and Combinations	Probability	Conditional Probability	Bayes Theorem
Probability Distributions	Poisson Distribution	Continuous Probability Distribution	Sampling
Non Probability Sampling	CLT	Hypothesis Testing	

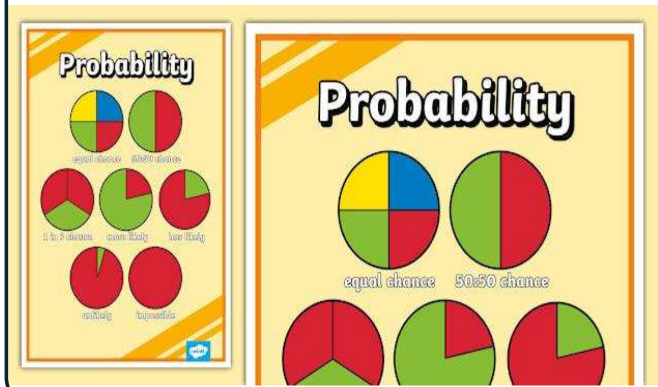
AI Mathematics Mindset

is all about *thinking about AI like and Engineer, Scientist and Decision Maker*

Q. Find the number of ways in which an ice cream sundae containing three flavors can be created out of a total of five flavors.

AI Mathematics Mindset – Probability Foundation

Probability is the measure of how likely an event is to happen



Q. Consider a scenario where a person has tested positive for an illness. This illness is known to impact about 1.2% of the population at any given time.

- The diagnostic test is known to have an accuracy of 85%, for people who have the illness. The test accuracy is 97% for people who do not have the illness.
- What is the probability that this person suffers from this illness, given that they have tested positive for it?

AI Mathematics Mindset – Statistical Thinking in AI

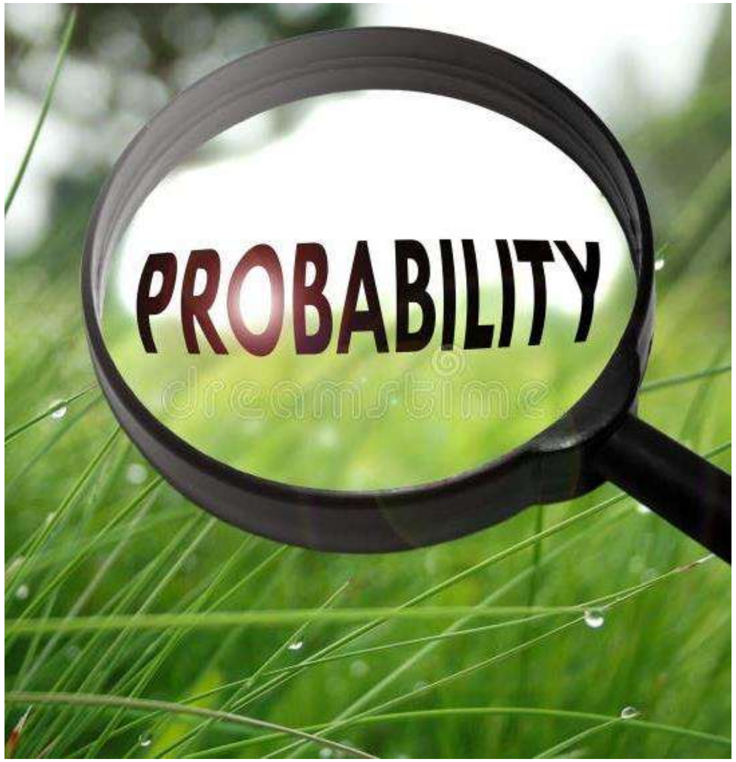
..is understanding data through patterns, probabilities, trends, distributions and uncertainty..

Q. What is the probability of a mail being spam, given that it contains the word “offer”?

- Available data indicates that 50% of all emails are spam mails. 9% of spam emails contain the word “offer,” and 0.4 % of ham emails contain the word “offer.”

AI Mathematics Mindset – Probability Foundation

- The metro rail company surveys eight senior citizens traveling in a subway train about their satisfaction with the new safety features introduced in the subway trains. Each response has only two values: yes or no.
- Let us assume that the probability of a "yes" response is 0.6, and the probability of a "no" response is 0.4 based on historical survey.
- Calculate the probability that :
- 1. Exactly three people are satisfied with the metro's new safety features
- 2. Fewer than five people are satisfied



AI Mathematics Mindset – Poisson Probability Problem

- In a subway station, the average number of ticket-vending machines out of operation is two. Assuming that the number of machines out of operation follows a Poisson distribution, calculate the probability that a given point in time:
- 1. Exactly three machines are out of operation
- 2. More than two machines are out of operation

AI Mathematics Mindset – Optimization and Decision Scoring Problem

- **Example 1 —** Weighted Decision Making in AI
- **Scenario**
- An AI system evaluates whether a loan should be approved.
- The system uses:
  - Income score
  - Credit score
  - Repayment history
- Each feature has different importance.

- **Example 2 — Why AI Needs Optimization**
- **Scenario**
- A house-price AI model predicts:
  - Actual price = 100
  - Predicted price = 130
- AI must reduce error.

Vectors, Embeddings & Semantic Space

- **Vectors**
- A **vector** is a list of numbers representing information mathematically.
- **Embeddings**
- An **embedding** is a special type of vector that captures semantic meaning.
- **Semantic Space**
- A **semantic space** is a mathematical space where embeddings are positioned based on meaning.
- In semantic space:
  - similar concepts stay close
  - unrelated concepts stay far apart

Matrices, Tensors & Transformer Operations

- **Matrices**
- A **matrix** is a rectangular arrangement of numbers organized into:
  - rows
  - columns
- It is essentially a collection of vectors.

- **Tensors**
- A **tensor** is a generalized multidimensional mathematical structure.
- Scalar = 0D tensor
- Vector = 1D tensor
- Matrix = 2D tensor
- Higher dimensions = tensors

Matrices, Tensors & Transformer Operations

- **Transformer Operations**
- Transformers are the mathematical foundation behind:
- ChatGPT
- Gemini
- Claude
- Copilot
- modern GenAI systems

Probability & Token Prediction

- Probability and token prediction form the core mathematical foundation of:
    - Large Language Models (LLMs)
    - ChatGPT
    - Gemini
    - Claude
    - Generative AI systems
- LLMs fundamentally work as:
    - **next-token probability prediction engines.**

Probability & Token Prediction - 1

- **Case Study 1 — ChatGPT Next-Word Prediction**
- **Scenario**
- A GenAI chatbot receives the sentence:
- “The capital of France is”
- The LLM computes probabilities for possible next tokens:

Token	Probability
Paris	0.91
London	0.05
Berlin	0.03
Apple	0.01

Probability & Token Prediction - 2

- **Question 1**
- Why did the model assign the highest probability to “Paris”?
- **Question 2**
- What mathematical concept powers this prediction?

Probability & Token Prediction - 1

Token	Probability
Infection	0.58
Pneumonia	0.27
Fracture	0.03
Allergy	0.12

- **Case Study 2 — AI Medical Assistant**
- **Scenario**
- A GenAI medical assistant receives:
- "Patient has high fever and persistent cough."
- Predicted next-token probabilities:

Probability & Token Prediction - 2

- **Question 1**
- Why are “infection” and “pneumonia” assigned higher probabilities?
- **Question 2**
- Why are low-temperature settings preferred in medical AI?

Calculus, Gradients and Learning

- Calculus and gradients are the mathematical mechanisms that allow AI systems to:
  - learn from mistakes
  - improve predictions
  - optimize behavior
  - train neural networks
  - power LLMs like OpenAI ChatGPT
- Without calculus:
  - modern AI cannot learn

Calculus, Gradients and Learning - 1

- **Case Study 1 — Training a Healthcare Diagnosis Model**
- **Scenario**
- A healthcare AI model predicts disease risk.
- The model equation is:
- $y=wx$

Calculus, Gradients and Learning - 2

- **Question 1**
- Calculate the predicted output.
- **Question 2**
- Calculate prediction error.
- **Question 3**
- Find Mean Squared Error (MSE).

Similarity Metrics & Vector Search

- Similarity metrics and vector search are core mathematical mechanisms behind:
  - semantic search
  - Retrieval-Augmented Generation (RAG)
  - recommendation systems
  - AI memory systems
  - AI agents
  - modern LLM retrieval pipelines

Similarity Metrics & Vector Search - 1

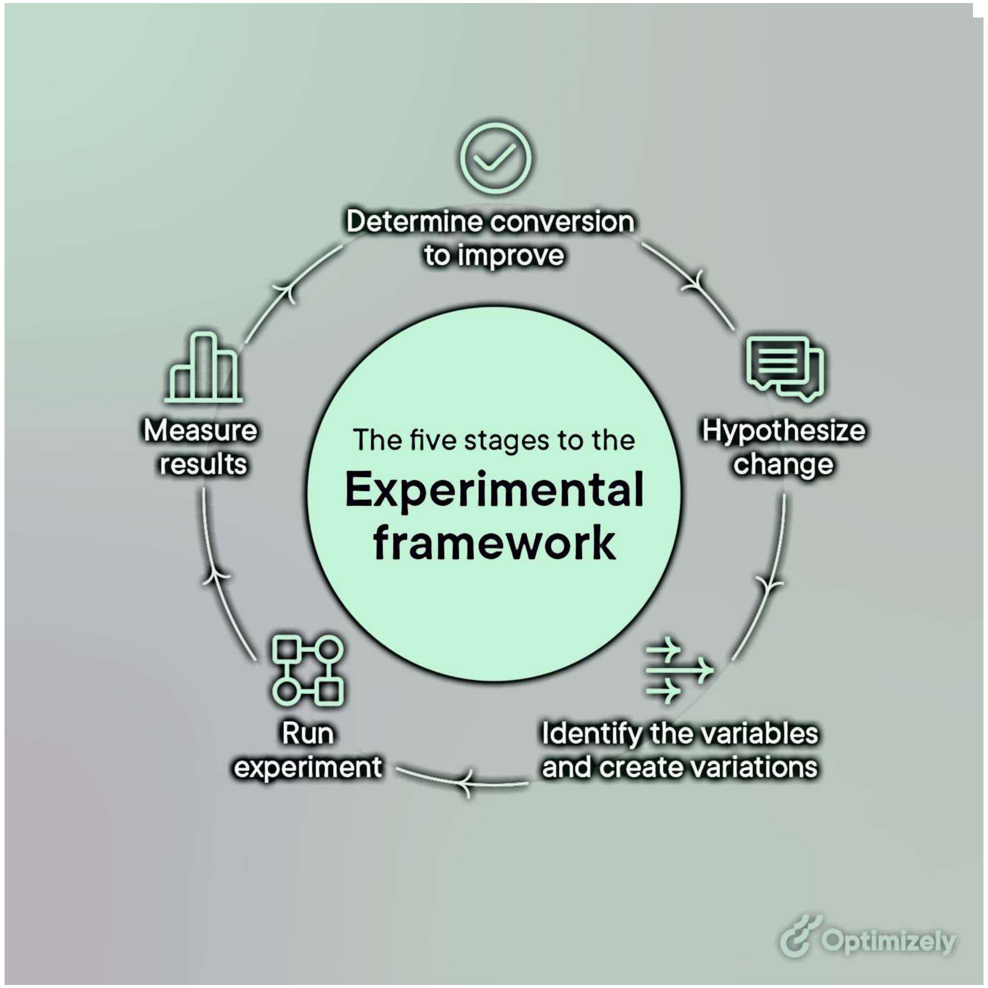
- **Case Study 1 — Healthcare Semantic Search System**
- **Scenario**
- A hospital deploys a GenAI-powered medical retrieval assistant.
- Doctor searches:
- “heart attack treatment”
- The system stores embeddings for medical documents.
- The system uses cosine similarity.

Similarity Metrics & Vector Search - 2

- **Question 1**
- Calculate cosine similarity between:
- Q and D1
- **Question 2**
- Which document should the RAG system retrieve?

Optimization & AI Agent Decisioning

- Optimization and decisioning are the core mechanisms that allow AI agents to:
    - choose actions
    - plan workflows
    - maximize success
    - minimize errors
    - improve performance autonomously
- Modern Agentic AI systems fundamentally work as:
- **optimization-driven decision systems.**



Optimization & AI Agent Decisioning - 1

- **Case Study 1 — AI Agent Tool Selection**
- **Scenario**
- An Agentic AI assistant receives the request:
- “Generate quarterly financial analysis.”
- The agent evaluates three tools:

Tool	Utility Score
Database Query	0.92
Web Search	0.45
Image Generator	0.08

- The agent must select the optimal action.

Optimization & AI Agent Decisioning - 2

- **Question 1**
- Which tool should the agent select?
- **Question 2**
- Why is optimization important in AI agent decisioning?

Optimization & AI Agent Decisioning - 1

Action	Reward
Correct retrieval	+15
Incorrect retrieval	-8
Infinite loop	-25

- **Case Study 2 — Reward Optimization in AI Agents**
- **Scenario**
- An autonomous research agent receives rewards:

- The agent performs:
- 2 correct retrievals
- 1 incorrect retrieval

Optimization & AI Agent Decisioning - 2

- **Question 1**
- Calculate total reward.
- **Question 2**
- What mathematical objective does reinforcement learning maximize?

Additional – Hands on

THANK YOU